

Redding's Closure Stability Conjecture

Closure-Generated Geometry (RDG Form — Canonical Block)

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Abstract

Dynamics obey a Fermat Principle: trajectories minimize a coherence-weighted action rather than Euclidean length. This variational perspective allows geometry to emerge from the closure flow Γ , with distances and geodesics determined by dynamical connectivity and coherence. It reverses the classical Riemannian paradigm in which a metric is prescribed a priori. The resulting closure-generated geometry aligns naturally with Lagrangian Coherent Structures, transfer-operator methods, and Conley-type invariants, offering a unified dynamical-first framework for coherent organization.

1 Notation and Setup

- P : primitive state space (normed vector space or complete metric space)
- $\Gamma = \{\Gamma_t\}_{t \geq 0}$: closure flow (continuous semigroup)
- $M \subseteq P$: invariant sets and manifolds
- μ : family of coherence measures
- $\lambda > 0$: decay parameter

2 Intuition and Analogy

Classical geometry begins with a manifold equipped with a metric, then studies dynamics upon it.

Closure-generated geometry inverts the order: the dynamics Γ *induces* the geometry. The effective structure arises from invariant sets, basins of attraction, coherence measures μ , and stability properties of Γ .

This perspective resonates with Lagrangian Coherent Structures (LCS), dynamic embeddings, and ergodic transport theory.

Fermat-Type Variational Principle

The starting point of the RDG framework is a Fermat-type principle for dynamical systems. Instead of minimizing Euclidean arclength, trajectories extremize a coherence-weighted action. Let $C_\Gamma(x, \dot{x})$ denote the coherence density induced by the closure flow Γ . The action functional is defined by

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

A trajectory $x(t)$ is said to obey the RDG Fermat Principle if it satisfies the variational condition

$$\delta S[x] = 0.$$

This principle asserts that coherent dynamical organization arises from extremizing a closure-induced action, providing the foundational variational structure from which closure-generated geometry emerges.

Coherence-Weighted Action

To formalize the variational principle, we introduce the coherence-weighted action functional. Let $C_\Gamma : TP \rightarrow [0, \infty)$ denote the coherence density induced by the closure flow $\Gamma = \{\Gamma_t\}_{t \geq 0}$, evaluated along a trajectory and its instantaneous velocity. For an absolutely continuous curve $x : [0, T] \rightarrow P$, define the action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

The quantity $C_\Gamma(x, \dot{x})$ plays the role of a refractive index: regions of high coherence correspond to regions of high optical density, causing trajectories to bend toward dynamically coherent structures.

The Fermat-type variational principle asserts that dynamically relevant trajectories are precisely those that extremize $S[x]$ under fixed endpoint constraints. Thus the action S is the fundamental object from which the closure-generated geometry is derived.

Closure Flow Γ

The closure flow is the dynamical engine from which coherence, action, and geometry are derived. Formally, a *closure flow* is a one-parameter semigroup

$$\Gamma = \{\Gamma_t\}_{t \geq 0}, \quad \Gamma_0 = \text{id}, \quad \Gamma_{t+s} = \Gamma_t \circ \Gamma_s,$$

acting on the primitive state space P . The flow may arise from a reduced model, a coarse-graining procedure, or an effective dynamical description that suppresses unresolved degrees of freedom.

The role of Γ in the RDG framework is twofold:

1. It determines the *dynamical connectivity* of the system, specifying which states evolve coherently under the induced dynamics.
2. It generates the *coherence density* $C_\Gamma(x, \dot{x})$ that weights the action functional and thereby shapes the emergent geometry.

In this sense, Γ plays the role of a “background medium” whose coherence properties determine how trajectories bend, cluster, or separate. The geometry that emerges from the variational principle is therefore not imposed externally but is *induced* by the closure dynamics encoded in Γ .

Emergent Geometry

The central inversion of the RDG framework is that geometry is not prescribed a priori. Instead, the closure flow Γ induces a coherence structure that determines how trajectories bend, cluster, and separate. This coherence structure, encoded in the density $C_\Gamma(x, \dot{x})$, acts as an effective refractive index that shapes the variational behavior of trajectories.

In classical Riemannian geometry, one begins with a metric tensor g and then studies geodesics and dynamics defined upon it. In contrast, RDG reverses this paradigm:

$$\text{dynamics} \implies \text{coherence} \implies \text{action} \implies \text{geometry}.$$

The geometry is therefore *closure-generated*: distances, neighborhoods, and distinguished directions arise from the variational structure of the action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

Regions of high coherence behave like regions of high optical density, causing trajectories to refract toward dynamically stable or invariant structures. Conversely, regions of low coherence behave like dynamically thin or dispersive regions, producing rapid divergence of nearby trajectories.

Thus the emergent geometry is not an external scaffold but a derived object reflecting the intrinsic organization of the closure dynamics. It is the natural geometric setting in which coherent structures, invariant manifolds, and transport barriers become geodesically meaningful.

RDG Distance

The coherence-weighted action induces a natural notion of distance on the primitive state space P . Given two states $A, B \in P$, the RDG distance is defined by the minimal coherence-weighted action among all admissible curves connecting them:

$$d_\Gamma(A, B) := \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

The infimum is taken over all absolutely continuous curves with fixed endpoints. A curve that attains this infimum, when it exists, is a closure geodesic.

This definition parallels the classical construction of geodesic distance in Riemannian geometry, but with a crucial difference: the integrand $C_\Gamma(x, \dot{x})$ is not derived from a prescribed metric tensor. Instead, it is induced by the closure flow Γ and its coherence structure. Thus the distance d_Γ is a *closure-generated* quantity, reflecting the dynamical organization of the system.

Regions of high coherence contribute more heavily to the action, effectively increasing the optical density of the medium. As a result, the RDG distance is sensitive to the dynamical pathways favored by Γ , and geodesics tend to align with coherent structures, invariant manifolds, and transport barriers.

Closure Geodesics

Given the RDG distance

$$d_\Gamma(A, B) = \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt,$$

a natural notion of geodesic arises from the variational structure of the action. A curve $\gamma : [0, T] \rightarrow P$ is called a *closure geodesic* if it locally minimizes the coherence-weighted action

$$S[\gamma] = \int_0^T C_\Gamma(\gamma(t), \dot{\gamma}(t)) dt$$

among all absolutely continuous curves with the same endpoints.

In contrast with classical Riemannian geometry, the integrand $C_\Gamma(x, \dot{x})$ is not derived from a fixed metric tensor but from the coherence structure induced by the closure flow Γ . As a

result, closure geodesics tend to align with dynamically coherent regions, invariant manifolds, and transport barriers, reflecting the intrinsic organization of the underlying dynamics.

When the infimum in the definition of $d_\Gamma(A, B)$ is attained, the minimizing curve is a closure geodesic. Even when minimizers do not exist globally, local minimizers still capture the geometric influence of coherence and provide the correct notion of dynamically preferred pathways in the closure-generated geometry.

Euler–Lagrange Equations

Closure geodesics arise as extremals of the coherence-weighted action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

Assuming C_Γ is continuously differentiable in both arguments, the first variation $\delta S[x]$ yields the generalized Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial C_\Gamma}{\partial \dot{x}}(x(t), \dot{x}(t)) \right) - \frac{\partial C_\Gamma}{\partial x}(x(t), \dot{x}(t)) = 0.$$

This equation characterizes the stationary curves of the action and therefore the closure geodesics. In contrast with classical mechanics or Riemannian geometry, the Lagrangian $C_\Gamma(x, \dot{x})$ is not derived from a kinetic energy or a metric tensor. Instead, it is induced by the coherence structure of the closure flow Γ .

The resulting Euler–Lagrange dynamics encode how coherence gradients and dynamical connectivity shape the preferred pathways of the system. Regions of high coherence act as regions of high optical density, bending extremal trajectories toward invariant manifolds, coherent sets, and dynamically stable structures. Thus the Euler–Lagrange equation provides the differential form of the closure-generated geometry.

3 Principle: Closure-Generated Geometry

The variational framework developed above leads naturally to a geometric interpretation of the closure dynamics. The coherence density $C_\Gamma(x, \dot{x})$ acts as an effective refractive index, and the action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

determines the preferred trajectories of the system through the Euler–Lagrange equations. The resulting extremals define closure geodesics, and the induced distance

$$d_\Gamma(A, B) = \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

provides a metric-like structure on the primitive state space P or on the quotient $P/\sim_\Gamma$ when distinct points share identical closure behavior.

In this sense, the geometry of the system is not imposed externally but is *generated* by the closure flow Γ . Invariant sets, their stable and unstable manifolds, and coherent regions of the flow appear as geodesically distinguished structures. The closure dynamics therefore endow P with a natural geometric organization, one that reflects dynamical connectivity, coherence, and long-term stability properties.

This motivates the following conjecture.

Dynamical Connectivity

The closure flow Γ induces a natural notion of dynamical connectivity on the primitive state space P . Two states are dynamically connected when their trajectories under Γ evolve coherently for a significant duration, as measured by the coherence density $C_\Gamma(x, \dot{x})$. Regions in which trajectories remain close, align in direction, or share long-term dynamical fate correspond to regions of high coherence.

This connectivity structure is reflected directly in the closure-generated geometry. States that evolve coherently incur low action cost, while states that diverge rapidly incur high cost. Thus the RDG distance

$$d_\Gamma(A, B) = \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

is small precisely when A and B lie within the same dynamically coherent region of the flow. Conversely, states separated by transport barriers, invariant manifolds, or regions of low coherence exhibit large RDG distance.

In this way, the closure flow Γ determines the geometric organization of P . Invariant sets, their stable and unstable manifolds, and coherent regions of the flow appear as geodesically distinguished structures. Dynamical connectivity is therefore the mechanism by which closure dynamics generate the effective geometry of the system.

Relation to Lagrangian Coherent Structures

The closure-generated geometry exhibits a natural alignment with Lagrangian Coherent Structures (LCS), though the two frameworks arise from distinct principles. LCS identify material surfaces that organize finite-time transport, typically through stretching-based diagnostics such as finite-time Lyapunov exponents or variational criteria for hyperbolic, elliptic, or parabolic structures. In contrast, RDG identifies geodesically distinguished structures through the coherence-weighted action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt,$$

where C_Γ encodes dynamical coherence rather than strain.

Despite these differences, the two perspectives often agree in coherent regions of the flow. High-coherence regions correspond to regions of low stretching, and transport barriers or invariant manifolds frequently appear as both LCS ridges and closure geodesics. In such regions, the RDG distance d_Γ becomes large across LCS barriers and small along coherent pathways, producing geodesics that align with the dominant transport skeleton of the system.

The key distinction is temporal: LCS are inherently finite-time objects, while the closure-generated geometry reflects the long-term or coarse-grained behavior encoded in Γ . As a result, RDG provides a geometric framework that persists beyond finite observation windows, while LCS capture transient transport features. When the closure flow accurately reflects the dominant dynamical organization, the two frameworks exhibit strong structural resonance.

Relation to Transfer-Operator Methods

Transfer-operator methods provide a spectral viewpoint on coherent behavior in dynamical systems. The Perron–Frobenius operator propagates densities forward in time, and its dominant eigenfunctions identify regions that resist mixing and maintain coherence over finite or infinite time horizons. These coherent sets correspond to almost-invariant regions of the flow and form the backbone of transport organization.

The RDG framework aligns naturally with this perspective. The coherence density $C_\Gamma(x, \dot{x})$ may be interpreted as a local proxy for the persistence of mass or probability under the closure flow Γ , analogous to the modulus of a dominant transfer-operator eigenfunction. Regions where the transfer operator exhibits slow decay or strong spectral concentration correspond to regions of high coherence, and therefore to regions of high optical density in the RDG variational principle.

As a consequence, closure geodesics tend to remain within almost-invariant sets and avoid regions of rapid mixing. The RDG distance d_Γ becomes small along pathways that follow the dominant spectral structures of the transfer operator and large across spectral gaps or mixing interfaces. This produces a geometric interpretation of coherent sets: they are regions in which the action cost is minimized and geodesics are naturally trapped.

Thus RDG provides a variational and geometric counterpart to the spectral viewpoint of transfer-operator theory. When the closure flow Γ captures the dominant transport features of the system, the two frameworks exhibit strong agreement, with coherent sets appearing simultaneously as spectral objects and as geodesically distinguished regions.

Relation to Conley Index Theory

Conley index theory provides a topological framework for understanding invariant sets, their isolating neighborhoods, and the qualitative structure of dynamical flows. An isolated invariant set S is characterized by an isolating neighborhood N such that all trajectories that remain in N for all time lie in S . The Conley index captures the essential topological features of S and remains invariant under perturbations of the flow.

The RDG framework aligns with this perspective by assigning geometric significance to invariant sets and their stable and unstable manifolds. Regions in which trajectories remain confined for long durations correspond to regions of high coherence density $C_\Gamma(x, \dot{x})$, and therefore to regions of high optical density in the variational principle. As a result, closure geodesics tend to remain within isolating neighborhoods and avoid their exit sets, reflecting the same dynamical boundaries that define the Conley index.

In this sense, the RDG distance d_Γ provides a geometric refinement of the topological structure captured by the Conley index. Distances become large across exit sets and small within invariant regions, producing geodesics that trace the stable and unstable manifolds associated with isolated invariant sets. The closure-generated geometry therefore recovers the qualitative organization emphasized by Conley theory, but enriches it with a quantitative, variational, and metric interpretation.

Thus RDG stands in natural correspondence with Conley index theory: both frameworks identify invariant structures and their dynamical boundaries, but RDG does so through a coherence-weighted action that endows these structures with geometric meaning.

Applications of Closure-Generated Geometry

The RDG framework provides a unified geometric perspective on coherent organization in dynamical systems, and its variational structure makes it applicable across a wide range of contexts. Because the geometry is induced by the closure flow Γ , the resulting distances, geodesics, and coherence structures adapt naturally to the dynamical regime under consideration. Several representative applications include:

- 1. Coherent Transport and Mixing.** In fluid dynamics and geophysical flows, closure geodesics identify pathways of minimal dispersion and highlight transport barriers. High-coherence regions correspond to low-action corridors, while mixing interfaces appear as regions of large RDG distance. This provides a geometric complement to LCS and transfer-operator diagnostics.

2. Model Reduction and Coarse-Graining. When Γ arises from a reduced or coarse-grained model, the RDG geometry reveals which degrees of freedom remain dynamically coherent under the reduction. Closure geodesics identify the effective low-dimensional manifolds along which the reduced dynamics faithfully represent the full system.

3. Stability and Invariant Set Detection. Invariant sets and their stable and unstable manifolds appear as geodesically distinguished structures. The RDG distance becomes small along invariant manifolds and large across their boundaries, providing a geometric method for detecting and characterizing stability regions.

4. Data-Driven Dynamics and Embeddings. When Γ is constructed from data (e.g., via dynamic mode decomposition, diffusion maps, or operator-learning methods), the closure-generated geometry provides a coordinate-free way to identify coherent sets, dominant modes, and long-term dynamical structure directly from observations.

5. Control and Optimal Intervention. Because the action $S[x]$ quantifies the coherence cost of a trajectory, it can be used to design control strategies that steer the system along or away from coherent pathways. Low-action directions correspond to dynamically efficient interventions, while high-action directions indicate regions where control is expensive or ineffective.

These applications illustrate the versatility of closure-generated geometry: it provides a variational, geometric, and dynamical framework that adapts to the structure of the closure flow and reveals the coherent organization underlying complex systems.

Structural Summary

The closure-generated geometry can be summarized as a sequence of transformations that convert dynamical information into geometric structure. Each stage introduces a new object—coherence density, action, geodesics, distance—that enriches the representation of the closure flow Γ .

$$\text{Closure Flow } \Gamma \xrightarrow{\text{induces}} C_\Gamma(x, \dot{x}) \xrightarrow{\text{weights}} S[x] = \int C_\Gamma dt \xrightarrow{\delta S=0} \text{Geodesics} \xrightarrow{\text{minimize}} d_\Gamma(A, B)$$

This schematic highlights the inversion at the heart of RDG:

$$\text{dynamics} \implies \text{coherence} \implies \text{action} \implies \text{geometry}.$$

The closure flow determines the coherence density, which determines the action, whose extremals define geodesics, which in turn induce a distance on P or on the quotient $P/\sim_\Gamma$. Invariant sets, coherent regions, and transport barriers appear as geodesically distinguished structures, providing a unified geometric interpretation of the dynamical organization encoded in Γ .

This structural summary serves as a conceptual map for the closure-generated geometry and motivates the conjecture that follows.

Axiomatic Form of the RDG Pipeline

The closure-generated geometry may be expressed as a sequence of axioms that formalize the transformation from dynamics to geometry. These axioms specify the minimal structures required for the RDG framework and clarify the logical dependencies among coherence, action, geodesics, and distance.

Axiom 1 (Closure Flow). There exists a continuous semigroup $\Gamma = \{\Gamma_t\}_{t \geq 0}$ acting on the primitive state space P , with $\Gamma_0 = \text{id}$ and $\Gamma_{t+s} = \Gamma_t \circ \Gamma_s$.

Axiom 2 (Coherence Density). The closure flow induces a coherence density

$$C_\Gamma : TP \rightarrow [0, \infty),$$

continuous in both arguments, representing the local coherence of the trajectory and its velocity under Γ .

Axiom 3 (Action Functional). For every absolutely continuous curve $x : [0, T] \rightarrow P$, the coherence-weighted action is defined by

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

Axiom 4 (Variational Principle). Dynamically relevant trajectories are extremals of the action:

$$\delta S[x] = 0.$$

When C_Γ is differentiable, extremals satisfy the Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial C_\Gamma}{\partial \dot{x}} \right) - \frac{\partial C_\Gamma}{\partial x} = 0.$$

Axiom 5 (Closure Geodesics). A curve γ is a closure geodesic if it locally minimizes $S[\gamma]$ among curves with the same endpoints. Extremals of the action are geodesics.

Axiom 6 (RDG Distance). The closure-generated distance between $A, B \in P$ is defined by

$$d_\Gamma(A, B) = \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

When minimizers exist, they are closure geodesics.

Axiom 7 (Induced Geometry). The action, geodesics, and distance together define a geometry on P (or on the quotient $P/\sim_\Gamma$ when distinct points share identical closure behavior). Invariant sets, coherent regions, and transport barriers appear as geodesically distinguished structures.

These axioms formalize the RDG inversion:

$$\Gamma \implies C_\Gamma \implies S[x] \implies \delta S = 0 \implies \text{geodesics} \implies d_\Gamma.$$

They provide the minimal structural foundation for the closure-generated geometry and motivate the conjecture that follows.

Minimal Example: Duffing Closure Geometry

To illustrate the closure-generated geometry in a concrete setting, consider the damped Duffing oscillator

$$\ddot{x} + \delta \dot{x} - x + x^3 = 0,$$

with $\delta > 0$. The system possesses two stable equilibria at $x = \pm 1$ and an unstable equilibrium at $x = 0$. The damping term ensures that trajectories converge toward one of the two wells, providing a natural closure flow Γ_t defined by the forward dynamics.

Coherence Structure. Trajectories within each potential well remain close for long durations and converge toward the same attractor. This produces a high coherence density C_Γ within each well and a low coherence density near the separatrix. The separatrix itself acts as a transport barrier: small perturbations on opposite sides lead to divergent long-term behavior.

Action and Geodesics. The coherence-weighted action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

assigns low cost to trajectories that remain within a single well and high cost to trajectories that cross the separatrix. As a result, closure geodesics tend to remain inside a well and avoid the unstable region near $x = 0$. The geodesics align with the dominant stable directions of the flow and reflect the basin geometry.

RDG Distance. The RDG distance $d_\Gamma(A, B)$ is small when A and B lie in the same well and large when they lie in different wells. In particular, points on opposite sides of the separatrix exhibit large RDG distance even when their Euclidean distance is small. This demonstrates the fundamental inversion of RDG: geometry is determined by dynamical fate, not spatial proximity.

Interpretation. This minimal example shows how closure-generated geometry identifies coherent regions, transport barriers, and invariant structures. The two wells appear as geodesically coherent regions, the separatrix appears as a region of high closure curvature, and the RDG distance reflects the dynamical organization of the system. Even in this simple setting, the closure flow induces a nontrivial geometry that cannot be captured by Euclidean distance alone.

Example: Double Gyre and Finite-Time Coherence

To illustrate closure-generated geometry in a finite-time, nonautonomous setting, consider the periodically forced double gyre flow

$$\dot{x} = -\pi A \sin(\pi f(x, t)) \cos(\pi y), \quad \dot{y} = \pi A \cos(\pi f(x, t)) \sin(\pi y) \frac{\partial f}{\partial x}(x, t),$$

where

$$f(x, t) = \varepsilon \sin(\omega t) x^2 + (1 - 2\varepsilon \sin(\omega t)) x.$$

This system exhibits two oscillating gyres separated by a time-dependent transport barrier. The periodic forcing induces stretching, folding, and mixing, making the double gyre a standard benchmark for coherent structure detection.

Closure Flow. A natural closure flow Γ_t is obtained by integrating the system over a finite time window $[0, T]$ and treating the resulting flow map as the effective evolution operator. This finite-time closure captures the dominant transport features without requiring long-term asymptotics.

Coherence Structure. Trajectories that remain within a single gyre over the interval $[0, T]$ exhibit high coherence, while trajectories near the oscillating separatrix experience strong stretching and mixing. The coherence density $C_\Gamma(x, \dot{x})$ therefore attains large values near the moving transport barrier and smaller values within each gyre.

Action and Geodesics. The coherence-weighted action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

assigns low cost to trajectories that remain inside a gyre and high cost to trajectories that cross the oscillating separatrix. Closure geodesics therefore trace the coherent cores of the gyres and avoid the mixing region. These geodesics align with the finite-time transport skeleton of the flow.

RDG Distance. The RDG distance $d_\Gamma(A, B)$ is small for points within the same gyre and large for points separated by the oscillating barrier. Even when Euclidean distance is small, points on opposite sides of the separatrix exhibit large RDG distance due to their distinct finite-time dynamical fates.

Interpretation. The double gyre example demonstrates that closure-generated geometry naturally captures finite-time coherent structures. The gyre cores appear as geodesically coherent regions, the oscillating separatrix appears as a region of high closure curvature, and the RDG distance reflects the finite-time transport organization of the system. This complements the long-term closure example of the Duffing oscillator and shows that RDG applies equally well to nonautonomous, mixing-dominated flows.

Example: Two-Attractor System and Quotient Geometry

To illustrate how closure-generated geometry induces a natural quotient structure on the state space, consider a system with two attracting sets A_1 and A_2 separated by a basin boundary. A canonical example is a planar gradient flow

$$\dot{x} = -\nabla V(x),$$

where V has two local minima and a saddle separating their basins of attraction. The closure flow Γ_t is given by the forward dynamics, which send each initial condition to one of the two attractors.

Closure Equivalence. Two points $x, y \in P$ are closure-equivalent if their trajectories converge to the same attractor:

$$x \sim_\Gamma y \iff \lim_{t \rightarrow \infty} \Gamma_t(x) = \lim_{t \rightarrow \infty} \Gamma_t(y).$$

Each equivalence class corresponds to a basin of attraction. The quotient space $P / \sim_\Gamma$ therefore consists of two points, representing the two attractors.

Coherence Structure. Within each basin, trajectories converge toward the same attractor and remain coherent for long durations. This produces a high coherence density C_Γ inside each basin and a sharp drop in coherence near the basin boundary. The boundary acts as a transport barrier: small perturbations across it lead to distinct long-term fates.

Action and Geodesics. The coherence-weighted action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

assigns low cost to trajectories that remain within a single basin and high cost to trajectories that cross the basin boundary. Closure geodesics therefore remain inside a basin unless forced by boundary conditions. The geodesics align with the stable directions of the gradient flow and avoid the saddle region.

RDG Distance and Quotient Geometry. The RDG distance $d_\Gamma(A, B)$ is small for points within the same basin and large for points in different basins. In particular, points arbitrarily close in Euclidean distance but lying on opposite sides of the basin boundary exhibit large RDG distance. This demonstrates that the closure-generated geometry naturally collapses each basin to a coherent region and inflates the boundary between basins.

Passing to the quotient $P/\sim_\Gamma$ identifies all points in a basin as a single geometric entity. The RDG distance between the two quotient points reflects the action cost of crossing the basin boundary and encodes the dynamical separation between the two attractors.

Interpretation. This example shows how closure-generated geometry induces a natural quotient structure on the state space. Basins of attraction appear as geodesically coherent regions, the basin boundary appears as a region of high closure curvature, and the RDG distance reflects the dynamical separation between attractors. The quotient geometry $P/\sim_\Gamma$ provides a coarse but dynamically faithful representation of the system.

Synthesis

The preceding development shows that the closure flow Γ induces a coherence density, an action functional, a family of geodesics, and a distance that together define a geometry reflecting the dynamical organization of the system. Across long-term, finite-time, and quotient-inducing examples, this geometry consistently identifies coherent regions, transport barriers, and invariant structures as geodesically distinguished. These observations motivate the following conjecture, which asserts that closure-generated geometry provides a faithful and intrinsic representation of the dynamical structure encoded by Γ .

4 Principle: Closure-Generated Geometry

Conjecture 1 (Closure-Generated Geometry). *Let $\Gamma = \{\Gamma_t\}_{t \geq 0}$ be a closure flow on a primitive state space P , and let $C_\Gamma : TP \rightarrow [0, \infty)$ be the coherence density induced by Γ . Let $S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$ be the associated action functional, and let d_Γ be the closure-generated distance obtained by minimizing $S[x]$ over curves with fixed endpoints.*

The conjecture is that the geometry induced by d_Γ faithfully represents the dynamical organization encoded by Γ . *More precisely:*

1. *Coherent regions of the flow correspond to geodesically coherent regions: points that share long-term or finite-time dynamical fate have small RDG distance.*

2. *Transport barriers, invariant manifolds, and mixing interfaces correspond to regions of large RDG distance and high closure curvature.*
3. *Invariant sets and their stable/unstable manifolds appear as geodesically distinguished structures: closure geodesics align with the dominant dynamical pathways of Γ .*
4. *The quotient space $P/\sim_\Gamma$, where $x \sim_\Gamma y$ if their trajectories share the same closure fate, inherits a well-defined geometry from d_Γ that reflects the coarse dynamical structure of the system.*
5. *The RDG geometry is stable under perturbations of Γ that preserve its coherent organization, and it agrees with established notions of coherence from LCS, transfer-operator theory, and Conley index theory whenever these frameworks identify robust dynamical structure.*

In summary, the conjecture asserts that the closure flow Γ generates a canonical geometry on P (or on $P/\sim_\Gamma$) and that this geometry provides an intrinsic, variational, and dynamically faithful representation of the system.

Discussion: Implications, Tests, and Failure Modes

The conjecture asserts that the closure-generated geometry induced by d_Γ faithfully reflects the dynamical organization of the system. If true, this provides a unified geometric framework for coherent structures, invariant sets, and transport barriers across a wide range of dynamical regimes. The conjecture therefore has both theoretical and practical implications.

Implications. If the conjecture holds, then the RDG distance provides a canonical metric for dynamical systems, one that captures long-term behavior, finite-time coherence, and basin structure in a single variational framework. Closure geodesics would offer a geometric tool for identifying coherent regions, invariant manifolds, and mixing interfaces, complementing and extending existing methods such as LCS, transfer-operator theory, and Conley index theory. The quotient geometry $P/\sim_\Gamma$ would provide a principled coarse-grained representation of the system.

Empirical Tests. The conjecture can be tested by comparing RDG geodesics and distances with known dynamical structures in benchmark systems. In particular:

- In systems with well-understood invariant manifolds, closure geodesics should align with stable and unstable directions.
- In mixing-dominated flows, RDG distances should distinguish coherent sets from rapidly mixing regions.
- In multi-attractor systems, the quotient geometry should collapse basins and inflate basin boundaries in accordance with dynamical fate.

Agreement across these regimes would provide strong evidence for the conjecture.

Failure Modes. The conjecture may fail if the closure flow Γ does not accurately capture the dominant dynamical organization of the system. In such cases, the coherence density C_Γ may misrepresent the true coherence structure, leading to geodesics and distances that do not align with invariant sets or transport barriers. Failure may also occur in systems with highly irregular or discontinuous closure behavior, where the variational principle does not produce meaningful geodesics or where minimizers do not exist.

Interpretation. These considerations highlight both the promise and the limitations of the closure-generated geometry. The conjecture provides a unifying hypothesis that connects variational, geometric, spectral, and topological perspectives on dynamical coherence. Its validity depends on the extent to which the closure flow Γ captures the essential dynamical structure of the system.

5 Prototype Closure Metric

A natural closure-generated trajectory metric arises by integrating the discounted separation of closure trajectories. For a closure flow Γ_t , define

$$d_\Gamma(x, y) = \int_0^\infty e^{-\lambda t} \|\Gamma_t(x) - \Gamma_t(y)\| dt, \quad \lambda > 0. \quad (1)$$

The exponential factor ensures that the metric is dominated by near-future coherence, reflecting the fact that closure geometry is determined by the portion of the trajectory where the closure flow retains predictive power. When trajectories remain close under Γ_t , the integrand remains small and $d_\Gamma(x, y)$ is small; rapid divergence produces large cost.

Under mild assumptions on Γ_t (continuity, subexponential growth), d_Γ defines a pseudo-metric on P . It becomes a true metric on the quotient space $P/\sim_\Gamma$, where $x \sim_\Gamma y$ if their closure trajectories coincide asymptotically. In this sense, d_Γ is the metric expression of closure equivalence: it collapses coherent regions and inflates dynamical boundaries.

This prototype metric is the simplest representative of the closure-generated geometry. More refined constructions replace the Euclidean separation with a coherence density $C_\Gamma(x, \dot{x})$, but the discounted trajectory metric captures the essential idea: geometry is induced by the dynamical fate of trajectories under the closure flow.

6 Formal Development — Core Structures

6.1 Closure Metric as Pseudo-Metric

Standing Hypotheses (H1–H4):

- (H1) $(P, \|\cdot\|)$ is a normed vector space or complete metric space.
- (H2) $\Gamma = \{\Gamma_t\}_{t \geq 0}$ is a continuous flow: $(t, x) \mapsto \Gamma_t(x)$ is jointly continuous.
- (H3) Growth bound: $\exists M > 0, \alpha \in \mathbb{R}$ such that $\|\Gamma_t(x) - \Gamma_t(y)\| \leq M e^{\alpha t} \|x - y\|$ for all $t \geq 0, x, y \in P$.
- (H4) $\lambda > \alpha$.

Theorem 2 (d_Γ is a Pseudo-Metric). *Under (H1–H4), d_Γ is a pseudo-metric on P :*

1. $d_\Gamma(x, y) \geq 0$,
2. $d_\Gamma(x, y) = d_\Gamma(y, x)$,
3. $d_\Gamma(x, z) \leq d_\Gamma(x, y) + d_\Gamma(y, z)$,
4. $d_\Gamma(x, x) = 0$.

Proof. (i) The integrand is non-negative. (ii) Follows from symmetry of the norm. (iii) Triangle inequality of the norm passes inside the integral. (iv) The integrand vanishes identically when $x = y$. $\square$

Theorem 3 (Convergence). *Under (H1–H4), $0 \leq d_\Gamma(x, y) \leq \frac{M}{\lambda - \alpha} \|x - y\| < \infty$.*

Proof. By (H3),

$$e^{-\lambda t} \|\Gamma_t(x) - \Gamma_t(y)\| \leq M \|x - y\| \exp((\alpha - \lambda)t).$$

The right-hand side integrates to $M \|x - y\| / (\lambda - \alpha) < \infty$ since $\lambda > \alpha$. $\square$

Theorem 4 (Identity of Indiscernibles (Tightened)). *$d_\Gamma(x, y) = 0$ if and only if $\|\Gamma_t(x) - \Gamma_t(y)\| = 0$ for Lebesgue-almost every $t \geq 0$.*

Moreover, if $t \mapsto \Gamma_t(x)$ and $t \mapsto \Gamma_t(y)$ are continuous, then $d_\Gamma(x, y) = 0$ implies $\Gamma_t(x) = \Gamma_t(y)$ for all $t \geq 0$.

Proof. If the set $\{t : \|\Gamma_t(x) - \Gamma_t(y)\| > 0\}$ has positive Lebesgue measure, then the integral is strictly positive since the integrand is nonnegative and continuous. The converse is immediate. Continuity upgrades almost-everywhere equality to everywhere equality. $\square$

Thus d_Γ is a true metric on P precisely when distinct points have distinguishable trajectories. In general, it induces a true metric on the quotient $P / \sim_\Gamma$.

6.2 Quotient Space $P / \sim_\Gamma$

Definition 1 (Closure Equivalence). $x \sim_\Gamma y \iff d_\Gamma(x, y) = 0$.

Theorem 5 ($\sim_\Gamma$ is an Equivalence Relation). *Reflexive, symmetric, and transitive (transitivity follows from the triangle inequality of d_Γ).*

Let $\pi : P \rightarrow P / \sim_\Gamma$ be the canonical projection, with $[x]_\Gamma = \pi(x)$.

Definition 2 (Quotient Metric). $\tilde{d}_\Gamma([x], [y]) := d_\Gamma(x, y)$.

Theorem 6 ($\tilde{d}_\Gamma$ is a Metric on $P / \sim_\Gamma$). (i) *Well-defined: if $x \sim_\Gamma x'$ and $y \sim_\Gamma y'$, then*

$$|d_\Gamma(x, y) - d_\Gamma(x', y')| \leq d_\Gamma(x, x') + d_\Gamma(y, y') = 0.$$

Hence $d_\Gamma(x, y) = d_\Gamma(x', y')$.

(ii) *$\tilde{d}_\Gamma$ satisfies all metric axioms.*

The induced quotient topology has basis

$$U_\varepsilon([x]) = \{[y] \in P / \sim_\Gamma : \tilde{d}_\Gamma([x], [y]) < \varepsilon\}.$$

Remark 1 (Topology Compatibility). *Under additional assumptions (e.g., the flow is expansive), the closure-generated topology is compatible with the original topology on P . In general it may be coarser when distinct points share the same asymptotic closure fate.*

6.3 Closure Geodesics

Definition 3 (Coherence Density). $\Phi_\Gamma : P \rightarrow [0, \infty)$ *continuous (e.g., local coherence score from μ , reciprocal Lyapunov exponent, or dominant transfer-operator eigenfunction value).*

Definition 4 (Closure Action Functional). *For a piecewise C^1 curve $\gamma : [a, b] \rightarrow P$,*

$$C_\Gamma(\gamma) := \int_a^b \Phi_\Gamma(\gamma(s)) \|\gamma'(s)\| ds.$$

Definition 5 (Closure Geodesic). *A curve γ is a closure geodesic if it locally minimizes C_Γ among admissible curves with the same endpoints in P or the same equivalence classes in $P / \sim_\Gamma$.*

Theorem 7 (Existence — Direct Method (Conditional)). *Suppose $(P/\sim_\Gamma, \tilde{d}_\Gamma)$ is compact (or proper and geodesically complete) and Φ_Γ is continuous and bounded from below. Then minimizing rectifiable curves exist between any two points in the quotient.*

Conjecture 8 (Geometric Characterization). *Closure geodesics are curves that remain inside maximal coherent sets or invariant manifolds for the longest possible duration, thereby minimizing cumulative divergence under Γ . Φ_Γ acts like a refractive index.*

Expected properties:

- Invariant manifolds are totally geodesic.
- Flow lines of Γ inside coherent regions are natural geodesics.
- Basin boundaries exhibit high closure curvature.

7 Future Work

Investigations

The closure-generated geometry framework suggests several directions for further investigation. These directions span theoretical development, computational methods, and applications to complex dynamical systems.

1. Existence and Regularity of Geodesics. A rigorous analysis of the variational problem underlying d_Γ is needed to establish conditions under which minimizers exist, are unique, or depend smoothly on their endpoints. This includes understanding degeneracies in C_Γ and the behavior of geodesics near transport barriers.

2. Characterization of Closure Curvature. The notion of closure curvature—regions where the action landscape exhibits sharp gradients—requires a precise geometric formulation. Developing curvature-like invariants for the RDG geometry may provide new tools for identifying coherent structures and transport barriers.

3. Numerical Methods for RDG Geodesics. Efficient algorithms for computing closure geodesics and RDG distances remain an open challenge. Variational integrators, graph-based approximations, and operator-learning approaches may offer promising computational strategies.

4. Data-Driven Construction of Γ . When the closure flow is inferred from data, questions arise regarding robustness, sampling requirements, and the effect of noise. Understanding how data-driven approximations of Γ influence C_Γ , $S[x]$, and d_Γ is essential for practical applications.

5. Connections to Spectral and Topological Methods. Further exploration of the relationship between RDG, transfer-operator theory, and Conley index theory may reveal deeper structural correspondences. In particular, it is unclear when spectral gaps or topological invariants guarantee the existence of geodesically coherent regions.

6. Applications to High-Dimensional Systems. Applying RDG to high-dimensional or infinite-dimensional systems—such as fluid flows, neural dynamics, or climate models—requires new techniques for representing and computing the closure flow. Understanding how RDG scales with dimension is a key open problem.

These directions highlight the breadth of the RDG program and the opportunities for advancing both the theoretical and computational foundations of closure-generated geometry.

Programs

The closure-generated geometry framework opens a broad research program spanning variational analysis, geometric structures, spectral methods, and data-driven approximation. Several directions appear particularly promising.

- 1. Analytical Foundations of RDG Geodesics.** A rigorous theory of existence, uniqueness, and regularity for closure geodesics remains to be developed. This includes understanding degeneracies in C_Γ , the structure of minimizers near transport barriers, and the behavior of geodesics under perturbations of the closure flow.
- 2. Closure Curvature and Geometric Invariants.** The notion of closure curvature—regions where the action landscape exhibits sharp gradients—suggests the possibility of curvature-like invariants intrinsic to the RDG geometry. Formalizing these invariants may provide new tools for identifying coherent structures and quantifying dynamical separation.
- 3. Computational Methods for RDG Distances.** Efficient algorithms for computing d_Γ and closure geodesics are essential for practical applications. Promising approaches include variational integrators, graph-based approximations, and operator-learning methods that approximate the action landscape directly from data.
- 4. Data-Driven Construction of Closure Flows.** When Γ is inferred from observations, questions arise regarding robustness, sampling requirements, and the effect of noise. Understanding how data-driven approximations of Γ influence C_Γ , $S[x]$, and d_Γ is crucial for applications in high-dimensional or partially observed systems.
- 5. Connections to Spectral and Topological Methods.** Further exploration of the relationship between RDG, transfer-operator theory, and Conley index theory may reveal deeper structural correspondences. In particular, it is unclear when spectral gaps or topological invariants guarantee the existence of geodesically coherent regions.
- 6. High-Dimensional and Infinite-Dimensional Systems.** Applying RDG to high-dimensional or infinite-dimensional systems—such as fluid flows, neural dynamics, or climate models—requires new techniques for representing and computing the closure flow. Understanding how RDG scales with dimension is a key open problem.

These directions outline a research program aimed at developing the theoretical, computational, and applied foundations of closure-generated geometry.

Open Problems

The closure-generated geometry framework raises several precise mathematical questions whose resolution would significantly advance the theory. The following problems outline core challenges at the interface of variational analysis, geometry, and dynamical systems.

Problem 1 (Existence of Minimizers). Determine conditions on the coherence density C_Γ under which the action minimization problem defining d_Γ admits a minimizer. In particular, characterize the role of degeneracies, discontinuities, and regions of high closure curvature.

Problem 2 (Regularity of Geodesics). Establish regularity properties of closure geodesics. When are minimizers smooth? When can geodesics exhibit corners or discontinuities in velocity? How does the geometry behave near transport barriers or invariant manifolds?

Problem 3 (Stability Under Perturbations). Analyze the stability of C_Γ , $S[x]$, and d_Γ under perturbations of the closure flow. Identify conditions under which the RDG geometry varies continuously with Γ and characterize possible bifurcations in geodesic structure.

Problem 4 (Closure Curvature). Develop a rigorous definition of closure curvature and determine how it relates to dynamical features such as invariant manifolds, spectral gaps, and mixing interfaces. Does closure curvature admit a tensorial or scalar representation?

Problem 5 (Spectral Correspondence). Identify conditions under which dominant eigenfunctions of the transfer operator correspond to geodesically coherent regions in the RDG geometry. Determine whether spectral gaps imply the existence of low-action corridors or geodesic trapping.

Problem 6 (Topological Correspondence). Clarify the relationship between RDG geometry and Conley index theory. Under what conditions do isolating neighborhoods correspond to geodesically coherent regions? Can the RDG distance detect changes in the Conley index?

Problem 7 (Quotient Geometry). Characterize the geometry of the quotient space $P/\sim_\Gamma$ induced by the RDG distance. Determine when the quotient is a metric space, when it is geodesic, and how its structure reflects the basin geometry of the underlying system.

Problem 8 (High-Dimensional Systems). Develop methods for computing or approximating d_Γ in high-dimensional or infinite-dimensional systems. Identify structural assumptions that allow dimensional reduction without destroying the RDG geometry.

Problem 9 (Data-Driven Approximation). When Γ is inferred from data, determine how sampling, noise, and model error affect the coherence density and the resulting geometry. Establish error bounds for data-driven approximations of d_Γ .

Problem 10 (Characterization of RDG Metrics). Classify the metrics that can arise as d_Γ for some closure flow Γ . Determine whether RDG metrics satisfy additional geometric or dynamical constraints beyond those of general Finsler or path metrics.

These problems define the mathematical core of the RDG research program and highlight the challenges involved in developing a complete theory of closure-generated geometry.

8 Conclusion

This is a Fermat’s Principle for Dynamics. Closure dynamics generates the geometry in which coherent structures live. These geodesics are not shortest paths in space; they are least-resistance paths through coherence.

The shift from “geometry first, then dynamics” to “dynamics first, then geometry” places the RDG framework (Relational, Emergent, Driven, Dynamics, Invariant, Neighborhood, Geometry) in natural alignment with modern dynamical systems theory, ergodic theory, and metric geometry. This conjecture is a new variational principle for dynamical systems.

Topology, metric structure, and geometric organization become secondary objects *induced* by closure dynamics.

Acknowledgments: This formulation synthesizes ideas from Lagrangian Coherent Structures, transfer operators, and invariant manifold theory into a unified dynamical-first perspective.

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9 Appendices

Appendix: Minimal Notation

This appendix summarizes the core notation used throughout the development of closure-generated geometry. The goal is to provide a concise reference without introducing additional structure beyond what is required for the RDG framework.

State Space. P denotes the primitive state space. Its tangent bundle is TP .

Closure Flow. $\Gamma = \{\Gamma_t\}_{t \geq 0}$ is the closure flow acting on P , with $\Gamma_0 = \text{id}$ and $\Gamma_{t+s} = \Gamma_t \circ \Gamma_s$.

Coherence Density. $C_\Gamma : TP \rightarrow [0, \infty)$ is the coherence density induced by Γ , assigning a local coherence cost to $(x, \dot{x})$.

Action Functional. For an absolutely continuous curve $x : [0, T] \rightarrow P$, the coherence-weighted action is

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

Geodesics. A curve γ is a closure geodesic if it locally minimizes $S[\gamma]$ among curves with fixed endpoints. Extremals satisfy $\delta S = 0$.

RDG Distance. The closure-generated distance between $A, B \in P$ is

$$d_\Gamma(A, B) = \inf_{\substack{x(0)=A \\ x(T)=B}} \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt.$$

Closure Equivalence. Two points are closure-equivalent if they share the same closure fate:

$$x \sim_\Gamma y \iff \lim_{t \rightarrow \infty} \Gamma_t(x) = \lim_{t \rightarrow \infty} \Gamma_t(y).$$

The quotient space is $P/\sim_\Gamma$.

Geometric Objects. Geodesically coherent regions, transport barriers, and invariant sets are interpreted through the geometry induced by d_Γ .

This notation provides the minimal vocabulary required to state the RDG axioms, examples, and conjecture without introducing unnecessary formal overhead.

Appendix: Variational Derivations

This appendix provides the variational derivations underlying the action functional, Euler–Lagrange equations, and geodesic structure of the closure-generated geometry. The goal is to make explicit the analytic foundations of the RDG framework.

First Variation of the Action. Let $x : [0, T] \rightarrow P$ be an absolutely continuous curve and let $\eta : [0, T] \rightarrow TP$ be a smooth variation with $\eta(0) = \eta(T) = 0$. Consider the perturbed curve $x_\varepsilon(t) = x(t) + \varepsilon\eta(t)$. The first variation of the action

$$S[x] = \int_0^T C_\Gamma(x(t), \dot{x}(t)) dt$$

is given by

$$\left. \frac{d}{d\varepsilon} S[x_\varepsilon] \right|_{\varepsilon=0} = \int_0^T \left(\frac{\partial C_\Gamma}{\partial x}(x, \dot{x}) \cdot \eta + \frac{\partial C_\Gamma}{\partial \dot{x}}(x, \dot{x}) \cdot \dot{\eta} \right) dt.$$

Integrating by parts yields

$$\delta S[x] = \int_0^T \left(\frac{\partial C_\Gamma}{\partial x} - \frac{d}{dt} \frac{\partial C_\Gamma}{\partial \dot{x}} \right) \cdot \eta(t) dt.$$

Euler–Lagrange Equation. A curve x is an extremal of S if and only if $\delta S[x] = 0$ for all variations η . This yields the Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial C_\Gamma}{\partial \dot{x}}(x, \dot{x}) \right) - \frac{\partial C_\Gamma}{\partial x}(x, \dot{x}) = 0.$$

Solutions of this equation are closure geodesics.

Geodesic Equation in Local Coordinates. In local coordinates (x^i) on P , write

$$C_\Gamma(x, \dot{x}) = C_\Gamma(x^i, \dot{x}^i).$$

Then the Euler–Lagrange equation becomes

$$\frac{d}{dt} \left(\frac{\partial C_\Gamma}{\partial \dot{x}^i} \right) - \frac{\partial C_\Gamma}{\partial x^i} = 0, \quad i = 1, \dots, \dim P.$$

When C_Γ is quadratic in $\dot{x}$, this reduces to a generalized geodesic equation with a Finsler-type connection.

Second Variation and Stability. The second variation of the action determines the stability of geodesics. For a variation η ,

$$\delta^2 S[x](\eta, \eta) = \int_0^T \left(\eta^\top \frac{\partial^2 C_\Gamma}{\partial x^2} \eta + 2\eta^\top \frac{\partial^2 C_\Gamma}{\partial x \partial \dot{x}} \dot{\eta} + \dot{\eta}^\top \frac{\partial^2 C_\Gamma}{\partial \dot{x}^2} \dot{\eta} \right) dt.$$

Positivity of the second variation characterizes local minimizers of S .

Regularity of Geodesics. Regularity of solutions to the Euler–Lagrange equation depends on the smoothness of C_Γ :

- If C_Γ is C^2 in both arguments, geodesics are C^2 .
- If C_Γ is only continuous in x and convex in $\dot{x}$, geodesics may exist but need not be smooth.
- Near regions of high closure curvature, geodesics may exhibit rapid changes in direction or velocity.

Boundary Behavior. If C_Γ becomes large near transport barriers or invariant manifolds, the action penalizes trajectories that cross these regions. As a result, geodesics tend to remain within coherent regions unless boundary conditions force otherwise.

These derivations provide the analytic foundation for the RDG geometry and clarify the variational structure underlying closure geodesics and the RDG distance.

Appendix: Quotient Geometry

This appendix develops the metric and geometric properties of the quotient space $P/\sim_\Gamma$ induced by the closure flow. The goal is to clarify how the closure-generated distance d_Γ behaves under the identification of points with identical closure fate.

Closure Equivalence. Recall that two points $x, y \in P$ are closure-equivalent if

$$x \sim_\Gamma y \iff \lim_{t \rightarrow \infty} \Gamma_t(x) = \lim_{t \rightarrow \infty} \Gamma_t(y).$$

Each equivalence class corresponds to a basin of attraction or, more generally, a region of identical closure behavior.

Quotient Space. The quotient space $P/\sim_\Gamma$ consists of equivalence classes $[x] = \{y \in P : y \sim_\Gamma x\}$. The natural projection $\pi : P \rightarrow P/\sim_\Gamma$ sends each point to its closure class.

Induced Distance. The RDG distance d_Γ induces a distance on the quotient by

$$\tilde{d}_\Gamma([x], [y]) = \inf_{u \in [x], v \in [y]} d_\Gamma(u, v).$$

This definition ensures that the quotient inherits the minimal separation between closure classes.

Metric Properties. The induced distance $\tilde{d}_\Gamma$ satisfies:

- *Nonnegativity:* $\tilde{d}_\Gamma([x], [y]) \geq 0$.
- *Symmetry:* $\tilde{d}_\Gamma([x], [y]) = \tilde{d}_\Gamma([y], [x])$.
- *Triangle inequality:* inherited from d_Γ .

The only nontrivial property is separation: $\tilde{d}_\Gamma([x], [y]) = 0$ if and only if $[x] = [y]$. This holds when d_Γ assigns positive cost to crossing basin boundaries.

Collapse of Coherent Regions. Within a closure class $[x]$, the RDG distance is typically small:

$$d_{\Gamma}(u, v) \approx 0 \quad \text{for } u, v \in [x].$$

Thus the quotient collapses each coherent region to a single geometric entity. This reflects the fact that points with identical closure fate are dynamically indistinguishable in the long-term limit.

Inflation of Basin Boundaries. If u and v lie in different closure classes, then any curve connecting them must cross a region of low coherence density. As a result,

$$d_{\Gamma}(u, v) \gg 1 \quad \text{for } u \not\sim_{\Gamma} v.$$

This inflates the separation between closure classes and highlights the dynamical significance of basin boundaries.

Geodesics in the Quotient. A geodesic in the quotient corresponds to a curve in P that minimizes the action subject to remaining within or transitioning between closure classes. If the action cost of crossing a boundary is large, geodesics tend to remain within a single class unless forced by boundary conditions.

Topology of the Quotient. The topology of $P/\sim_{\Gamma}$ depends on the structure of closure classes:

- If closure classes are open, the quotient is discrete.
- If closure classes have nontrivial boundaries, the quotient may inherit a non-Hausdorff topology.
- If closure classes form a stratification, the quotient may be a stratified metric space.

Interpretation. The quotient geometry provides a coarse but dynamically faithful representation of the system. It collapses coherent regions, inflates transport barriers, and encodes the basin structure of the closure flow. The induced metric $\tilde{d}_{\Gamma}$ captures the minimal action required to transition between closure classes.

This analysis clarifies the geometric and metric structure of the quotient space $P/\sim_{\Gamma}$ and its role in the RDG framework.

Appendix: Computational Methods for RDG

This appendix outlines computational strategies for approximating the coherence density C_{Γ} , the action functional $S[x]$, closure geodesics, and the RDG distance d_{Γ} . These methods are essential for applying closure-generated geometry to high-dimensional or data-driven systems.

1. Graph-Based Approximations. When the state space is sampled at discrete points $\{x_i\}$, one may approximate the action by constructing a weighted graph whose edges represent local transitions. The edge weight between x_i and x_j is given by a discrete approximation of $C_{\Gamma}(x_i, x_j - x_i)$, and the RDG distance is approximated by the shortest path distance on this graph. This approach is robust to noise and scales well with data size.

2. Variational Solvers. For systems with smooth dynamics, closure geodesics may be computed by solving the Euler–Lagrange equations numerically. Methods include:

- shooting methods for boundary-value problems,
- collocation methods using polynomial or spline bases,
- variational integrators that preserve the structure of the action.

These solvers provide high accuracy but may be sensitive to initial guesses and nonconvexity in the action landscape.

3. Discrete Action Minimization. An alternative approach is to discretize the curve $x(t)$ into a sequence of points $\{x_k\}$ and minimize a discrete action

$$S_{\text{disc}}(x_0, \dots, x_N) = \sum_{k=0}^{N-1} C_{\Gamma}(x_k, x_{k+1} - x_k).$$

This reduces the geodesic problem to a finite-dimensional optimization problem, which can be solved using gradient-based or gradient-free methods. This approach is flexible and well-suited to high-dimensional systems.

4. Operator-Learning Approaches. When the closure flow Γ is inferred from data, one may approximate C_{Γ} using operator-learning techniques such as:

- neural operators,
- kernel-based regression,
- diffusion maps or manifold learning.

These methods approximate the action landscape directly from observations and can handle systems where the underlying dynamics are unknown or partially observed.

5. Approximation of the Closure Flow. In practice, the closure flow Γ may be approximated by:

- finite-time flow maps,
- ensemble averages of trajectories,
- transfer-operator approximations,
- data-driven surrogate models.

The accuracy of C_{Γ} , $S[x]$, and d_{Γ} depends on the fidelity of this approximation.

6. Numerical Stability and Regularization. Regions of high closure curvature may lead to numerical instability in geodesic computation. Regularization strategies include:

- smoothing the coherence density,
- penalizing rapid changes in velocity,
- adaptive discretization near transport barriers.

These techniques improve robustness without altering the underlying geometry.

7. High-Dimensional Systems. For high-dimensional systems, dimensional reduction techniques may be applied before computing RDG quantities. Methods include:

- principal component analysis,
- autoencoders,
- diffusion maps,
- Koopman embeddings.

The challenge is to preserve the structure of the closure flow under reduction.

These computational methods provide practical tools for approximating the RDG geometry in both model-based and data-driven settings. They complement the theoretical framework and enable applications to complex dynamical systems.

Appendix: RDG vs LCS (Formal Correspondence)

This appendix clarifies the relationship between closure-generated geometry and the finite-time Lagrangian Coherent Structure (LCS) framework. While LCS identifies transport barriers using finite-time deformation, RDG provides a variational geometry that generalizes these ideas and applies to a broader class of closure flows.

LCS Background. In the LCS framework, coherent structures are identified using the finite-time Cauchy–Green strain tensor

$$C_{t_0}^T(x) = (D\Phi_{t_0}^T(x))^\top D\Phi_{t_0}^T(x),$$

where $\Phi_{t_0}^T$ is the flow map over the interval $[t_0, t_0 + T]$. Hyperbolic LCS are typically defined as ridges of the largest eigenvalue of $C_{t_0}^T$.

Correspondence via Coherence Density. In RDG, the coherence density $C_\Gamma(x, \dot{x})$ plays the role of a generalized strain measure. When Γ is taken to be the finite-time flow map $\Phi_{t_0}^T$, one may define

$$C_\Gamma(x, \dot{x}) \approx \sqrt{\dot{x}^\top C_{t_0}^T(x) \dot{x}},$$

which recovers the LCS strain functional. In this case, closure geodesics coincide with curves of extremal stretching or contraction, and the RDG distance reflects finite-time deformation.

Transport Barriers. Hyperbolic LCS correspond to regions where the action landscape exhibits sharp gradients. In RDG terms, these are regions of high closure curvature, where geodesics avoid crossing due to the large action cost. Thus LCS ridges appear as geodesically repelling structures in the RDG geometry.

Coherent Sets. Elliptic LCS (vortex boundaries) correspond to regions where the action is nearly constant along closed curves. In RDG, these appear as geodesically coherent regions: trajectories within the region have small RDG distance, while trajectories outside incur a large action cost to enter.

Generalization Beyond LCS. RDG extends the LCS framework in several ways:

- It applies to closure flows that are not derived from physical trajectories.
- It incorporates long-term behavior, not just finite-time deformation.
- It provides a variational metric structure rather than a tensor field.
- It naturally handles quotient geometry and basin collapse.

Thus LCS appear as a special case of RDG when the closure flow is a finite-time flow map and the coherence density is derived from the Cauchy–Green tensor.

Interpretation. The correspondence between RDG and LCS shows that closure-generated geometry recovers classical finite-time coherent structures while providing a broader geometric framework. RDG unifies variational, spectral, and topological perspectives on coherence, with LCS emerging as a specific instantiation of the general theory.

Appendix: RDG vs Transfer Operators (Spectral Correspondence)

This appendix clarifies the relationship between closure-generated geometry and transfer-operator methods for identifying coherent sets. While transfer operators use spectral information to detect slowly mixing regions, RDG provides a variational geometry that captures similar structures through geodesics and action minimization.

Transfer Operator Background. Given a flow map $\Phi_t : P \rightarrow P$, the Perron–Frobenius operator $\mathcal{P}_t : L^2(P) \rightarrow L^2(P)$ is defined by

$$(\mathcal{P}_t f)(x) = f(\Phi_{-t}(x)).$$

Coherent sets correspond to almost-invariant sets under $\mathcal{P}_t$, typically identified using the leading nontrivial eigenfunctions of $\mathcal{P}_t$ or its normalized variants.

Spectral Coherence. If $\lambda_1 = 1 > \lambda_2 \geq \lambda_3 \geq \dots$ are the eigenvalues of $\mathcal{P}_t$, then a spectral gap

$$\lambda_2 \ll 1$$

indicates the presence of a coherent set. The corresponding eigenfunction partitions the state space into regions of slow mixing.

Correspondence via Coherence Density. In RDG, the coherence density C_Γ encodes the local resistance to mixing. When Γ is taken to be the finite-time flow map Φ_t , one may define

$$C_\Gamma(x, \dot{x}) \approx \text{local mixing rate at } x,$$

which is inversely related to the magnitude of the leading eigenfunctions of $\mathcal{P}_t$. Regions where the eigenfunctions vary slowly correspond to regions of low coherence cost and therefore small RDG distance.

Geodesic Interpretation of Spectral Gaps. A spectral gap in $\mathcal{P}_t$ implies the existence of a region A such that trajectories starting in A remain in A with high probability. In RDG terms, this means:

- C_Γ is small inside A ,
- closure geodesics remain inside A ,
- d_Γ is small for points in A ,
- d_Γ is large for points crossing ∂A .

Thus spectral coherence corresponds to geodesic coherence.

Action-Minimizing Sets vs Almost-Invariant Sets. Almost-invariant sets maximize retention under $\mathcal{P}_t$. In RDG, coherent regions minimize the action required to remain within the region. These two notions coincide when:

- the closure flow is derived from the physical flow,
- C_Γ reflects local mixing rates,
- the action penalizes transitions across mixing interfaces.

Generalization Beyond Spectral Methods. RDG extends transfer-operator methods in several ways:

- It applies to closure flows not derived from physical trajectories.
- It provides a geometric metric structure rather than a spectral partition.
- It handles long-term, finite-time, and quotient-induced coherence uniformly.
- It identifies geodesic pathways, not just coherent regions.

Thus transfer-operator coherence appears as a special case of RDG when the closure flow is a finite-time flow map and the coherence density reflects local mixing rates.

Interpretation. The correspondence between RDG and transfer operators shows that closure-generated geometry recovers spectral coherent sets while providing a broader variational and geometric framework. RDG unifies spectral, variational, and topological perspectives on coherence, with transfer-operator methods emerging as a specific instantiation of the general theory.

Appendix: RDG vs Conley Index (Topological Correspondence)

This appendix clarifies the relationship between closure-generated geometry and the Conley index, a topological invariant used to detect isolated invariant sets and their dynamical structure. While the Conley index captures topological persistence of invariant sets, RDG provides a variational geometry that identifies these sets through geodesic behavior and action minimization.

Conley Index Background. Given a continuous flow Φ_t on a state space P , an isolating neighborhood N is a compact set such that the maximal invariant set

$$\text{Inv}(N) = \{x \in N : \Phi_t(x) \in N \text{ for all } t \in \mathbb{R}\}$$

lies in the interior of N . The Conley index of $\text{Inv}(N)$ is defined using an index pair (N, L) and encodes the topological structure of the invariant set.

Closure Flow Interpretation. In RDG, the closure flow Γ may differ from the physical flow Φ_t , but when Γ is derived from Φ_t , closure classes often align with invariant sets or their stable/unstable manifolds. In particular:

- If N is an isolating neighborhood for Φ_t , then C_Γ is typically small inside N and large near ∂N .
- Closure geodesics remain inside N unless forced to exit.
- The RDG distance collapses $\text{Inv}(N)$ to a geodesically coherent region.

Thus isolating neighborhoods correspond to regions of low action in the RDG geometry.

Invariant Sets as Geodesically Distinguished Structures. If $S = \text{Inv}(N)$ is an isolated invariant set, then:

- d_Γ is small for points in S ,
- d_Γ is large for points crossing ∂N ,
- closure geodesics tend to align with stable/unstable manifolds of S .

This mirrors the Conley index perspective, where S is detected by its isolating neighborhood and exit set.

Correspondence via Quotient Geometry. The Conley index collapses the exit set L to a point. In RDG, the quotient geometry $P/\sim_\Gamma$ collapses entire closure classes, including invariant sets and their basins. When Γ reflects the physical flow, these two collapses coincide:

- Conley collapse: $L \mapsto *$,
- RDG collapse: $[x] \mapsto \text{closure class}$.

Both constructions identify dynamically equivalent points.

Detection of Morse Decompositions. A Morse decomposition is a finite collection of isolated invariant sets ordered by gradient-like flow. In RDG:

- each Morse set corresponds to a geodesically coherent region,
- the partial order corresponds to low-action pathways between regions,
- the RDG distance encodes the “energy barriers” between Morse sets.

Thus RDG provides a geometric realization of the Morse graph.

Generalization Beyond Conley Theory. RDG extends the Conley index in several ways:

- It applies to closure flows not derived from physical trajectories.
- It provides a metric structure, not just a topological invariant.
- It identifies geodesic pathways between invariant sets.
- It handles finite-time and coarse-grained dynamics naturally.

Conley theory appears as a special case when the closure flow is the physical flow and the coherence density reflects the isolating structure of invariant sets.

Interpretation. The correspondence between RDG and the Conley index shows that closure-generated geometry recovers topological invariants of dynamical systems while providing a richer variational and geometric framework. RDG unifies topological, spectral, and variational perspectives on coherence, with the Conley index emerging as a specific instantiation of the general theory.

Appendix: Further Derivations

This appendix provides the formal variational derivations underlying the closure-generated geometry framework. We derive the Euler–Lagrange equations for closure geodesics, discuss regularity and existence issues, and record the structural identities needed for later geometric development.

A. Closure Action and Admissible Paths

Let (P, Γ_t) be a primitive state space equipped with a closure flow $\Gamma_t : P \rightarrow P$ and coherence density

$$C : TP \rightarrow \mathbb{R}_{\geq 0}, \quad (x, \dot{x}) \mapsto C(x, \dot{x}).$$

For any absolutely continuous path $x : [0, T] \rightarrow P$, the closure action is

$$S[x] = \int_0^T C(x(t), \dot{x}(t)) dt.$$

Admissible paths satisfy:

- $x(0) = A, x(T) = B$ for prescribed endpoints,
- $C(x, \dot{x})$ finite a.e.,
- $\dot{x}$ measurable and locally integrable.

B. First Variation and Euler–Lagrange Equations

Let $x_\varepsilon(t) = x(t) + \varepsilon \eta(t)$ with $\eta(0) = \eta(T) = 0$. Then

$$\left. \frac{d}{d\varepsilon} S[x_\varepsilon] \right|_{\varepsilon=0} = \int_0^T \left(\frac{\partial C}{\partial x}(x, \dot{x}) \cdot \eta + \frac{\partial C}{\partial \dot{x}}(x, \dot{x}) \cdot \dot{\eta} \right) dt.$$

Integrating by parts yields

$$\delta S[x; \eta] = \int_0^T \left(\frac{\partial C}{\partial x} - \frac{d}{dt} \frac{\partial C}{\partial \dot{x}} \right) \cdot \eta dt.$$

Thus x is a closure geodesic iff it satisfies the Euler–Lagrange system

$$\frac{d}{dt} \left(\frac{\partial C}{\partial \dot{x}} \right) - \frac{\partial C}{\partial x} = 0.$$

C. Regularity and Existence

Assume:

- C is C^2 in $(x, \dot{x})$,
- C is strictly convex in $\dot{x}$,

- $C(x, \dot{x}) \geq \alpha \|\dot{x}\|^2 - \beta$ for some $\alpha > 0$.

Then:

1. Minimizers exist by the direct method of the calculus of variations.
2. Minimizers satisfy the Euler–Lagrange equations in the weak sense.
3. Under convexity and smoothness, minimizers are C^2 .

These conditions hold for all coherence densities of the form

$$C(x, \dot{x}) = w(x) \|\dot{x}\|^2,$$

with $w(x) > 0$ smooth, which includes all coherence-weighted Riemannian examples and many data-driven constructions.

D. Closure Momentum and Hamiltonian Form

Define the closure momentum

$$p = \frac{\partial C}{\partial \dot{x}}(x, \dot{x}).$$

If the Legendre transform is invertible, define the Hamiltonian

$$H(x, p) = p \cdot \dot{x} - C(x, \dot{x}),$$

with $\dot{x}$ expressed in terms of (x, p) .

Then closure geodesics satisfy Hamilton’s equations:

$$\dot{x} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial x}.$$

This form is useful for numerical integration and for relating RDG geodesics to invariant sets of the closure flow.

E. Second Variation and Stability

Let x be a geodesic and η a variation field with fixed endpoints. The second variation is

$$\delta^2 S[x; \eta] = \int_0^T \left(\eta^\top \frac{\partial^2 C}{\partial x^2} \eta + 2\eta^\top \frac{\partial^2 C}{\partial x \partial \dot{x}} \dot{\eta} + \dot{\eta}^\top \frac{\partial^2 C}{\partial \dot{x}^2} \dot{\eta} \right) dt.$$

Positivity of the second variation characterizes local minimality and determines the stability of geodesics under perturbations of endpoints or coherence density.

F. Structural Identity: Closure Curvature

Define the closure curvature operator $\mathcal{K}$ along a geodesic $x(t)$ by the Jacobi-type equation

$$\ddot{\eta} + \mathcal{K}(t)\eta = 0,$$

where $\mathcal{K}$ is derived from the second variation tensor. Regions of high coherence correspond to positive curvature, causing geodesics to bend toward coherent structures.

G. Summary

The closure action admits a complete variational formulation: existence, Euler–Lagrange equations, Hamiltonian form, second variation, and curvature. These structures justify interpreting closure geodesics as the intrinsic geometric skeleton of the dynamical system.